Shereen Anand

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Research Interests

Mathematics, Stochastic Modeling, Reinforcement Learning, Post-Training, Autonomous Agents, LLM Evaluation, Multimodal Learning, Computational Neuroscience, Optimization, Game Theory, and Natural Language Processing.

Education

Bachelor of Technology in Computer Science and Engineering **Sept 2022 – Jul 2026**
PES University, Bangalore *CGPA: 3.56/5.0*

KSEAB (12th Grade) **June 2020 - June 2022**
RV PU College *Score: 95.3%*

Experience

Research Sciences Intern **Oct 2026 – Present**
Microsoft Research (Manager: Gopal Srinivasan) *Bangalore, India*

- Evaluating the **coding agents** that run inside Copilot, identifying the **hidden test cases** that agent-written code fails but existing test suites miss because they do not exercise every behaviour of the code, and grouping them by failure type.

Applied Scientist Intern, Central Machine Learning Team **Jan 2026 – Jun 2026**
Amazon (Manager: Kaveri Sivaramakrishnan) *Bangalore, India*

- **Developed ASPEX**, an aspect-grounded persona generation framework integrating hierarchical taxonomy induction with soft assignment, and prompt optimization OPRO technique to model multi-faceted user preferences for personalized conversational recommendation.
- Designed and benchmarked persona-conditioned recommendation across **8 downstream personalization tasks** spanning **Amazon and 3 public datasets** with a cumulative of **300K+ users**, enforcing strict temporal separation between persona construction and downstream evaluation.
- Established state-of-the-art performance over contemporary persona generation baselines, achieving relative **24.2%** PR-AUC and **15.8%** ROC-AUC improvement in the engagement prediction. Observed **18.9%** higher recommendation follow-up rate in suggestions provided.
- Optimized production LLM inference on AWS Bedrock using Claude models, reducing end-to-end latency by 20.3% using an aspect-grounded verification methodology; work submitted to the **EACL Industry Track 2027**.

Student Researcher, Autonomy and Quality for Agents **Jun 2025 – Dec 2025**
Google DeepMind (Manager: Sagar Gubbi) *Bangalore, India*

- **Architected VISTA-Bench**, an adversarial evaluation framework for frontier agentic AI that auto-generates distributionally diverse testing environments using a dynamic Symmetric Adversarial Protocol.
- Engineered scalable evaluation pipelines across **300+ synthetic mobile environments**; formalized automated telemetry metrics for high-dimensional compositional generalization, behavioral trajectory consistency, and multi-step error recovery dynamics, minimizing empirical evaluation overhead by **35%**.
- Identified core brittleness modes in state-of-the-art frontier agents, systematically characterizing performance degradation profiles to prove that despite achieving **90%+ standard benchmark accuracy**, agent policy success rates decayed by up to **42%** under non-stationary covariate distribution shifts.
- Discovered and quantified a multilingual robustness paradox, establishing a statistically rigorous counter-intuitive phenomenon where frontier agents manifest a **22% increase** in out-of-distribution generalization vectors when evaluated using low-resource languages compared to high-resource baselines.

Research Experience

Research Collaborator **University of Calgary**
DENOS Lab, Advisor: Prof. Steve Drew *Calgary, Canada (Remote)*

- Extending **EDSim**, an LLM-agent emergency department simulator where GPT-4o-mini-driven patient, triage nurse, bedside nurse and physician agents run the full care journey, toward an ICML 2027 submission.
- Evaluating simulated wait times against historical data from a Level-1 trauma centre, stratified across **5 CTAS acuity levels**, with baseline outputs matching historical wait-time distributions.
- Running what-if surge experiments over 326 simulated patient visits (187 baseline, 139 surge) with a 60% arrival-rate increase, quantifying a **75.4% rise** in mean ED length of stay (552 to 968 minutes).
- Analyzing stage-level bottlenecks across a 79-bed, 15-physician ED replica, showing arrival-to-assessment times up **74.7%** and post-disposition boarding up **100.7%** under stress.

Research Intern **IIIT Bangalore**
Advisor: Prof. V. Ramasubramanian, Summer Research Internship Program *Bangalore, India*

- Constructed low-resource multilingual ASR systems leveraging **matching networks** and **self-supervised learning** to address data scarcity in underserved linguistic communities.
- Engineered cross-lingual transfer mechanisms using **wav2vec 2.0** with alignment-aware modeling, reducing **phoneme error rates by 28%** in complex code-switching scenarios.
- Analyzed phonetic confusion patterns across **15 language pairs**, identifying systematic misalignments in acoustic feature representations that propagate under extreme data scarcity (<**100 hours** labeled audio).
- Demonstrated that **self-supervised pretraining** can democratize speech technology access for **11 low-resource languages**, establishing baselines for future language-specific acoustic modeling.

Invited Talks

Anand, S. (2026). Can We Trust Agents With Tasks Behind a Login? In: *1st IEEE Workshop on Networked AI Agents in a Decentralized Architecture (NANDA)*, MIT Media Lab, co-located with IEEE TPS 2026, San Jose, CA, November 4-6, 2026.

Publications

Anand, S., Vamshi, V., Peddi, V., Prabhakaran, V. (2026). Enhanced Trading Systems with Intelligent Market Analysis. In: *Proceedings of the 4th International Conference on Artificial Intelligence and Applications (ICAIA)*. **Accepted (Main Track)**.[Paper]

Anand, S., Mourya, A., Mukerjee, R., Puranik, V., Kaveri, S. (2027). ASPEX: Aspect-Grounded Persona Generation with Hierarchical Taxonomy. In: *EACL Industry Track 2027*. **Under Review**. [OpenReview]

Projects

Deforestation Detection & Forecasting from Satellite Imagery | *Python, TensorFlow, GDAL, U-Net* **2024**

- Aggregated and preprocessed multi-temporal Sentinel-2 and Landsat imagery across **12 districts**, applying cloud-masking and NDVI-based filtering to construct a clean, geo-referenced training corpus.
- Trained a **U-Net segmentation model** to delineate forest cover loss at pixel level, reaching **89% IoU** on held-out regions through iterative tuning of encoder depth and loss weighting for class imbalance.
- Formulated an LSTM-based forecasting layer atop segmentation outputs to project deforestation trends 6 months ahead, cutting mean prediction error by **21%** against a linear-regression trend baseline evaluated on the same held-out districts.

Skin Cancer Detection using Deep Learning | *Python, PyTorch, CNN* **2023**

- Curated and preprocessed a dermoscopic image dataset of **10K+ lesion samples**, applying targeted augmentation (rotation, color-jitter, oversampling) to correct class imbalance in minority melanoma cases.
- Fine-tuned a **CNN classifier** (EfficientNet backbone) via transfer learning to distinguish malignant from benign lesions, attaining **92% classification accuracy** and **0.90 F1-score** on a stratified test split.
- Applied **Grad-CAM-based interpretability** to visualize activation regions driving predictions, cross-validating model attention against dermatologist-annotated lesion boundaries to check for spurious correlation.

Mean-Field Game Model of LLM Agent Teams | *Python, AutoGen, Optimisation* **2026**

- Proposed a **mean-field game** model of multi-agent LLM teams, treating each agent as facing the population share of agents in each role, to replace the many-agent problem with one representative-agent problem.
- Designed a validation plan comparing the real **AutoGen** team against the mean-field prediction for team sizes **N = 2, 4, 8, 16**, with the prediction error measured by total variation distance over matched runs.
- Planned an **equilibrium** test in which one agent deviates to its best alternative strategy, reporting the score gain as a lower bound on how exploitable the team is.

Teaching & Mentorship

Student Mentor **IEEE RAS**
PES University *2024–2025*

- Guided junior students through robotics competition projects and provided technical mentorship. Conducted workshops on robot programming.

Teaching Assistant **Practical Applied Machine Learning**
Centre for Entrepreneurship and Innovation, PES University *2024–2025*

- Designed lab exercises and practical sessions for 56 students covering ML concepts. Led hands-on tutorials implementing ML algorithms provided one-on-one guidance on project development and debugging.

Technical Skills

Programming Languages: Python, C++, R, MATLAB, SQL, Java, JavaScript, TypeScript, HTML/CSS
Machine Learning & AI: TensorFlow, PyTorch, scikit-learn, Keras, XGBoost, LightGBM, reinforcement learning, neural networks, ensemble methods, feature selection, hyperparameter optimization
Data Engineering: Apache Spark, Hadoop, Airflow, Pandas, NumPy, SciPy, real-time streaming, event processing, data quality checks, feature stores
Visualization & Analytics: Matplotlib, Seaborn, Plotly, Tableau, Power BI, interactive dashboards
Backend Development: Python, C++, Java, microservices, REST APIs, GraphQL, message queues
Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS, SQS), GCP (BigQuery, Dataflow), Azure, Docker, Kubernetes, Terraform, real-time data pipelines
Databases: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, InfluxDB, KDB+/q, time-series databases
System Design: High-frequency trading infrastructure, low-latency systems, distributed computing, microservices, load balancing, caching, fault tolerance
Development Tools: Git, GitHub, VSCode, IntelliJ, Jupyter, Docker, CI/CD, testing frameworks (Pytest, Jest), profiling
Operating Systems: Linux, Windows, macOS, Shell scripting, system administration

Achievements & Awards

1. Runner-Up – Amazon Internal Hackathon 2026
2. Top-100 – Google Girl Hackathon 2025 Software Round
3. Finalist – Smart India Hackathon in Software Edition 2023, selected at national level.
4. Runner-Up – Robowars Tournament in 2023
5. Finalist – Srinivasa Ramanujan Mathematics Competition (SRMC) 2016, invited to offline finals at IIT Bombay.